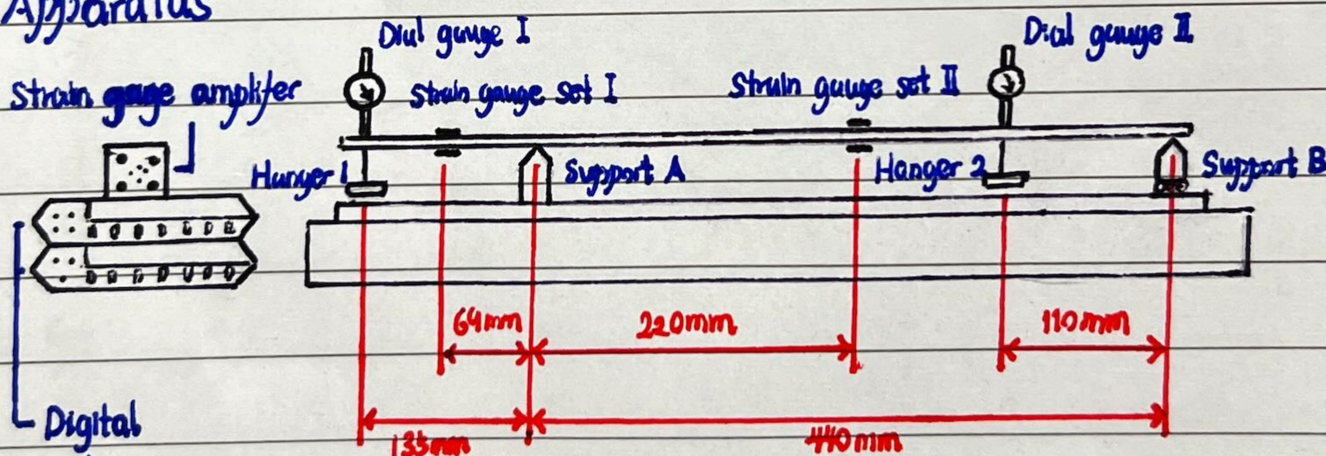
3/8/2023 Simple Beams Pre-Lab

Aiming

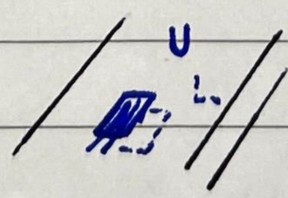
- 👁 behaviour of simple beam structures
  application of linear superposition
- ✓ response of a statically indeterminate beam
  → superposition of two statically determinate beams
- 👁 sensitivity of statically determina/indeterminate beams
  → change the height of a support
- compare experiment with theory

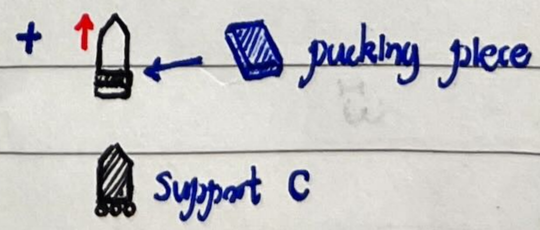
Apparatus



w = 18.86mm, t = 2.94mm, E = 200GPa

* strain gauge





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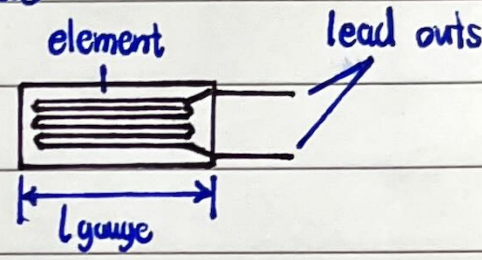
Procedure

- Reset equipment
- Connect the strain gauge, zero the SGA output in '~~200mV~~ 2V' scale

According to the Strain Measurement Pre-Lab

$R = \rho \frac{L}{A}$

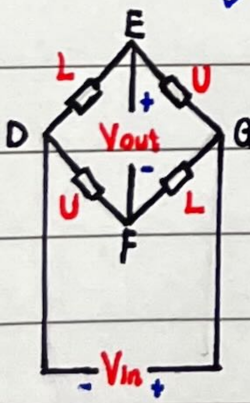


If ε is not too large

$\frac{\Delta R}{R} = S\varepsilon$

gauge factor = 2.12

measured by Wheatbridge since it is very small



$V_s = 1V$, GAIN = 500

$V_{out} = V_{in}\left[\frac{\Delta U}{U} - \frac{\Delta L}{L} + \frac{\Delta U}{U} - \frac{\Delta L}{L}\right]$

$\bar{V}_{out} = GAIN \cdot V_{out}$

$\varepsilon = \frac{\frac{\Delta R}{R}}{S} = \frac{V_{out}}{V_{in} S} = \frac{\bar{V}_{out}}{V_{in} S \cdot GAIN}$

If $\bar{V}_{out} = 1mV$, $V_{\cancel{in}}^{out} =$

① Apply load to Hanger 1

→ 20N at beginning 5N × 4

→ remove 5N each time

→ record reading

* '200mV' scale might switched as load rises

② Apply load to Hanger 2

→ 40N at beginning 5N × 8

3/8/2023 Simple Beams Pre-Lab

→ remove 5N each time

→ record reading

③ Apply load to both Hangers

→ 5N to 1 → 10N to 2

→ record reading

×4

→ 5N×4 at 1 → 10N×4 at 2

④ Lift support A

→ insert packing piece at A

→ repeat ①

→ remove packing piece and reset support

⑤ Statically indetermine condition

→ replace Hanger 1 by Support C

→ 40N at Hanger 2 at beginning 5N×8

→ remove 5N each time

→ record reading

⑥ Lift support A in statically indetermine condition

→ repeat ④

→ repeat ⑤

Prepare an excel sheet to record the data

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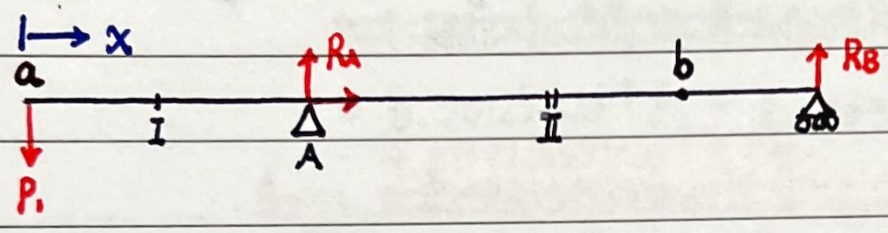
Prediction by theory

* ↻ ↓ positive

* calculate in m

① Apply load to Hanger 1



$P_1 + R_A + R_B = 0$

(B), $440 R_A + 575 P_1 = 0$ $\quad \therefore R_A = \frac{115}{88} P_1$ $\quad \therefore R_B = -\frac{27}{88} P_1$

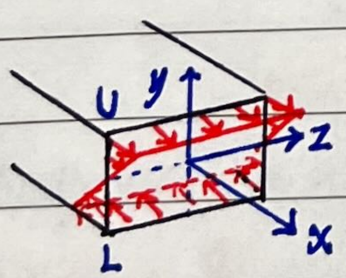
$\therefore M = P_1 x - \frac{115}{88} P_1 \langle x - 0.135 \rangle$

$\therefore M_I =$

$M_{II} =$

$M_a =$

$M_b =$



$M_y = 0$, $M_z = M$, $\sigma_x = \frac{M_z}{I_z} y = \varepsilon_x E$

$I_z =$

$\therefore \varepsilon = \frac{M}{I_z E} y$

$\therefore \varepsilon_{U,I} =$

$\varepsilon_{L,I} =$

$\varepsilon_{U,II} =$

$\varepsilon_{L,II} =$

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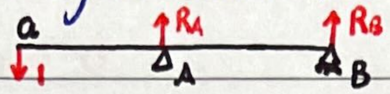
At a,



$\therefore \bar{R}_A = \frac{115}{88}$, $\bar{R}_B = -\frac{27}{88}$

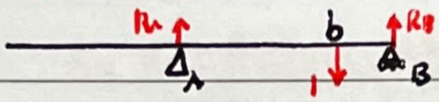
$\therefore \bar{M} = x - \frac{115}{88} \langle x - 0.135 \rangle$

$\because 1 \cdot \delta = \int_0^L \frac{M}{EI} \bar{M}\, dx$

$\therefore \delta E I_z = \int_0^{0.135} P_1 x^2\, dx + \int_{0.135}^{0.575} P_1 \left[x - \frac{115}{88} \langle x - 0.135 \rangle\right]^2 dx$

$\therefore \delta_a =$

At b,



$\therefore \bar{R}_A = \frac{1}{4}$, $\bar{R}_B = \frac{3}{4}$

$\therefore \bar{M} = -\frac{1}{4} \langle x - 0.135 \rangle + \langle x - 0.465 \rangle$

$\cancel{\delta EI = \int_{0.135}^{0.465} P_1 x \left[-\frac{1}{4} \langle x - 0.135 \rangle\right] dx +}$

$\cancel{+ \int_{0.465}^{0.465} P_1}$

$\therefore \delta E I_z =$

$=$

$=$

$\therefore \delta_b =$

② Apply load to Hanger 2

Similar to ①

$R_A = \frac{1}{4} P_2$, $R_B = \frac{3}{4} P_2$, $M = -\frac{1}{4} P_2 \langle x - 0.135 \rangle + P_2 \langle x - 0.465 \rangle$

$\therefore M_z =$

$M_a =$

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$\therefore \varepsilon_{U,I} =$

$\varepsilon_{L,I} =$

$\varepsilon_{U,II} =$

$\varepsilon_{L,II} =$

At a, $\delta EI_z = \int_{0.135}^{0.465} P_2 \left[\cancel{K} x - \frac{115}{88}(x-0.135)\right]\left(-\frac{1}{4}(x-0.135)\right] dx$

$+ \int_{0.465}^{0.575} P_2 \left[\left(x - \frac{115}{88}(x-0.135)\right)\left(-\frac{1}{4}(x-0.135) + (x-0.465)\right] dx$

$=$

$=$

$\therefore \delta a =$

At b, $\delta EI_z = \int_{0.135}^{0.465} P_2 \left(-\frac{1}{4}(x-0.135)\right)^2 dx$

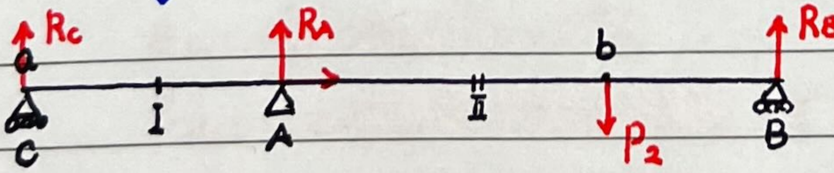
$+ \int_{0.465}^{0.575} P_2 \left[-\frac{1}{4}(x-0.135) + (x-0.465)\right]^2 dx$

$=$

$=$

$\therefore \delta b =$

③ statically indeterminte condition



$\mapsto x$ (C), $-135 R_A + 465 P_2 - 575 R_B = -27 R_A + 93 P_2 - 115 R_B = 0$

(B), $110 P_2 - 440 R_A - 575 R_C = 22 P_2 - 88 R_A - 115 R_C = 0$

$R_C + R_A + R_B = P_2$

$M = EIv'' = -R_C x - R_A \langle x - 0.135 \rangle + P_2 \langle x - 0.465 \rangle$

$\therefore EIv' = -\frac{1}{2} R_C x^2 - \frac{1}{2} R_A \langle x - 0.135 \rangle^2 + \frac{1}{2} P_2 \langle x - 0.465 \rangle^2 + C_1$

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$\therefore EIv^{x} = -\frac{1}{6} R_C x^3 - \frac{1}{6} R_A \langle x-0.135\rangle^3 + \frac{1}{6} P_2 \langle x-0.465\rangle^3 + C_1 x + C_2$

When $x=0$, $v=0$ $\therefore C_2 = 0$

When $x=0.575$, $v=0$

$\therefore -575^3 R_C - 440^3 R_A + 110^3 P_2 + 3450 C_1 = 0$

When $x=0.135$, $v=0$

$\therefore -135^3 R_C + 810 \cancel{x} C_1 = 0$

$\therefore -174630000 R_C - 85184000 R_A + 1331000 P_2 = 0$

$\therefore R_A =$

$R_B =$

$R_C = -$

$\therefore M =$

$\therefore M_I =$

$M_{II} =$

$\therefore \varepsilon_{U,I} =$

$\varepsilon_{L,I} =$

$\varepsilon_{U,II} =$

$\varepsilon_{L,II} =$

When $x = 0.465$,

$EI_2 v =$

$\therefore \delta_{II}(\downarrow) =$

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- Correct SGA output from 200 mV to 2V scale
- Is the relationship in Wheatstone bridge vaild ?

  $\varepsilon = \bar{V}_{out} \cdot \left(\frac{1}{1mV} \cdot 0.943\mu \right)$

  * lineur ?

  ** Excited V_s is seperute from the voltuge that measured
- When will the approximation by theory invaild ?
  * Exceed the small angle approximation
- Why 6 tests corried out instead of 1 ?
- Any errors/ uncertuinties need to be considered ?
- When insert support C, the beam displace according to the deflection gauge !

- Error might occurs, remember to recording

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			Load Hanger 1 (N)	Load Hanger 2 (N)	Strain Gauge 1 (V)	Strain Gauge 2 (V)	Deflection Gauge 1 (mm)	Deflection Gauge 2 (mm)
Uncertainty								
Error								
↓ Hanger 1	Test 1	Loading Hanger 1						
		Unloading Hanger 1						
↓ Hanger 2	Test 2	Loading Hanger 2						
		Unloading Hanger 2						
↓ both hunger ↓	Test 3	Simultaneous Loading						
		Simultaneous Unloading						
↓ packing ↓ A ⇧	Test 4	Insert packing A and Simultaneous Loading						
		Insert packing A and Simultaneous Unloading						

Statically determine condition

2X

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		Load Hanger 1 (N)	Load Hanger 2 (N)	Strain Gauge 1 (V)	Strain Gauge 2 (V)	Deflection Gauge 1 (mm)	Deflection Gauge 2 (mm)
Uncertainty							
Error							
Test 5	Loading Hanger 2 ↓ 2						
	Unloading Hanger 2 ↓ 2						
Test 6	Insert packing A and Loading Hanger 2 ↓ 2 + A						
	Insert packing A and Unloading Hanger 2 ↓ 2 + A						

Statically indetermine condition

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Prediction by theory

* Test 1. 2. 5 already done in Pre-Lab

* For M, ↻ ↺ positive

For ε, extension is positive

For δ, downward ↓ positive

① P_1 at Hanger 1

$M = P_1 \langle x - \frac{115}{88} P_1 \langle x - 0.135 \rangle \rangle$

$M_I =$

$M_{II} =$

$\varepsilon_{v,I}$

$\varepsilon_{v,II}$

$\delta_a =$

$\delta_b =$

② P_2 at Hanger 2

$M = P_2 \left(-\frac{1}{4} \langle x - 135 \rangle + \langle x - 465 \rangle \right)$

$M_I =$

$M_{II} =$

$\varepsilon_{v,I}$

$\varepsilon_{v,II}$

$\delta_a =$

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$\delta_b = 1.249703222 \times 10^{-1} P_2$

③ P_1 at Hanger 1, P_2 at Hanger 2

$\varepsilon_{v,I} =$

$\varepsilon_{v,II} =$

$\delta_a =$

$\delta_b =$

④ Lift support A then ③

$\varepsilon_{v,I} =$

$\varepsilon_{v,II} =$

$\delta_a =$

$\delta_b =$

⑤ Statically indeterminate

$M = \frac{121}{144} P_2 x - \frac{91}{144} \langle x - 135 \rangle P_2 + \langle x - 465 \rangle P_2$

$M_I =$

$M_{II} =$

$\varepsilon_{v,I} =$

$\varepsilon_{v,II} =$

$\delta_b =$

⑥ Lift support A then ⑤

ε changes

δ changes

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				Experiment Data								Prediction by theory			
		Load Hanger 1 (N)	Load Hanger 2 (N)	Strain 1	Absolute error	Strain 2	Absolute error	Deflection a (mm)	Absolute error	Deflection b (mm)	Absolute error	Strain 1	Strain 2	Deflection a (mm)	Deflection b (mm)
Test 1	Loading Hanger 1														
	Unloading Hanger 1														
Test 2	Loading Hanger 2														
	Unloading Hanger 2														
Test 3	Simultaneous Loading														
	Simultaneous Unloading														
Test 4	Insert packing A and Simultaneous Loading														
	Insert packing A and Simultaneous Unloading														

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Strain

Strain

Test 1

Test 2

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d_{II}

d_I

ε_{II}

ε_I

Test 3

Test 4

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		Experiment Data										Prediction by theory		
		Load Hanger 1 (N)	Load Hanger 2 (N)	Strain 1	Absolute error	Strain 2	Absolute error	Deflection a (mm)	Absolut error	Deflection b (mm)	Absolute error	Strain 1	Strain 2	Deflection b (mm)
Test 5	Loading Hanger 2													
	Unloading Hanger 2													
Test 6	Insert packing A and Loading Hanger 2													
	Insert packing A and Unloading Hanger 2													

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d_{II}

d_{I}

ε_{II}

ε_{I}

Test 5

Test 6

~~6~~ 7

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Validation Plot matches theory from experiments ✓

Effect of Raising Support see previous page (page 7 Pre-Lab)

statically determinate strain don't change, displacement change

statically indeterminate strain and displacement change

Validation of superposition

work for statically determinate beam (Test 3 = Test 1 + Test 2)

Error analysis

- see during-lab parts
- some possible source of error might occurs
  - disturbances on the 工作台 workbench as people lean/hit it
  - uneven loading and sudden jolts to the equipment when loading and unloading the weights may lead errors
  - ☆ small angle approximation becomes invalid at higher loads
  - ☆ hysteresis 迟滞现象
    (which can be observed from the data)
  - imperfections in the cross-section of the beam from material cause the different E across the beam

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Further improvement

- If can, consider if the incremental merit justify the effort
  i.e. consider the error from wearing of the weights

Addition points

About hysteresis:

a. if there are any valid sources of hysteresis:
   1. beam strain a lot;
   2. repeating loading leads fatigue;
   3. loading and unloading;

b. at what relative proportions? that their effects are.